

AI-Powered Question Generator

System Design & Feature Overview

PRODUCT BRIEF

v1.0

2026

A complete AI-powered exam preparation platform that lets students and teachers generate, customise, and download professional question papers in seconds — drawing from trusted knowledge bases and aligned to real exam-board syllabuses.

30+ Questions per set	8 Knowledge bases	5 Difficulty levels	7 Question types
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1 The Problem We're Solving

Students preparing for GCSE and A-Level exams spend hours searching for practice questions across multiple websites — PMT, Save My Exams, Corbettmaths, BBC Bitesize — only to find questions that are either too easy, too hard, or poorly matched to their specific exam board. Teachers face the same struggle when creating worksheets.

■ Time wasted	Manually hunting across 5+ sites for relevant practice questions takes 30–60 minutes per session.
■ Misalignment	Generic questions rarely match the exact specification of AQA, Edexcel, OCR, or CIE papers.
■ No mark schemes	Found questions often lack model answers or examiner-style mark schemes for self-assessment.
■ No customisation	You can't control difficulty, question count, or mix of types in one place.

2 The Solution

The AI Question Generator is a single, unified platform where you type a topic name, choose your level and preferences, and receive a fully formatted question paper with mark scheme — ready to download as PDF — in under 10 seconds.

Core idea

One input → AI generates exam-quality questions → instant PDF with mark scheme. No more tab-switching, no more mismatched difficulty, no more blank mark schemes.

Topic & Subject Input

Enter any topic name (e.g. 'Quadratic Equations', 'Photosynthesis', 'The Cold War'). Select the subject from a dropdown and specify the exam board — AQA, Edexcel, OCR, CIE, WJEC, or General.

Difficulty Levels

Five calibrated levels: Foundation, GCSE, AS Level, A Level, and University. Each level maps to real exam mark-scheme complexity, not just surface vocabulary.

Question Count Control

A slider lets you request 1–30 questions per set. Ideal for quick topic checks (5 Qs) or full timed mock papers (25–30 Qs).

Question Type Mix

Choose from 7 types: Multiple Choice, Short Answer, Long Answer / Essay, True/False, Fill in the Blank, Calculation, and Diagram / Label. Select any combination.

Instant Mark Scheme

Every question ships with a concise model answer. Toggle answers per question or reveal all at once. The PDF separates questions and mark scheme on different pages.

PDF Export

Download a professionally formatted PDF with the question paper on page 1 and the complete mark scheme on page 2 — including subject, board, level, total marks, and date.

Knowledge Bases

The AI draws on the style, structure, and question patterns from leading UK exam preparation resources. Users can toggle individual sources on or off for each session.

SOURCE	FOCUS AREA	BEST FOR
PMT (Physics & Maths Tutor)	Maths, Sciences, A Level	Structured past-paper style Qs
Save My Exams	All GCSE & A Level subjects	Concise, mark-scheme-aligned Qs
Corbettmaths	Mathematics (GCSE)	Worked-example & practice Qs
Khan Academy	All subjects, multi-level	Conceptual & foundational Qs
BBC Bitesize	All GCSE subjects	Accessible, curriculum-mapped Qs
Seneca Learning	GCSE & A Level	Retrieval practice style Qs
Past Papers (Official)	All subjects & boards	Authentic exam-question style
General AI	Any topic	Novel Qs beyond existing databases

Suggested additions

Mathway (step-by-step maths), GCSEPod (video-aligned revision), Isaac Physics (Cambridge-linked), Quizlet (community flashcard sets), and direct exam-board specimen papers (AQA, Edexcel, OCR portals).

5 Suggested Improvements

Content & Intelligence

- Syllabus mapping — tag each question to a specific syllabus point (e.g. AQA Biology 4.2.1) so revision is targeted.
- Adaptive difficulty — if a student answers correctly, the next question auto-scales harder (practice mode).
- Diagram / figure prompts — flag questions that typically require a drawn diagram or labelled graph.
- Formula sheet generator — auto-attach a relevant formula sheet for science/maths papers.

Workflow & Usability

- Topic mix mode — combine multiple topics in one paper (e.g. Algebra + Trigonometry + Statistics).
- Timed mock mode — full exam simulation with countdown timer and auto-submit.
- Question bank — save generated sets to a personal library and remix them later.
- Score yourself mode — tick correct/incorrect per question with auto-calculated score and percentage.

Export & Sharing

- Word (.docx) export — teachers can edit questions before printing.
- Teacher vs student PDF — same content, two versions: one with mark scheme hidden.
- QR code on PDF — links back to the digital interactive version.
- Google Classroom / Teams integration — share directly to a class with one click.

Analytics & Learning

- Performance tracker — log which topics a student struggles with over time.
- Spaced repetition — automatically resurface weak topics at the optimal review interval.
- Progress dashboard — visual charts showing mastery per topic and subject.
- AI feedback — when a student gets a question wrong, the AI explains why and suggests follow-up Qs.

6 How It Works

1

Input

Student or teacher enters a topic name, selects subject, exam board, difficulty level, question count, question types, and preferred knowledge bases.

2

Prompt Construction

The system assembles a structured prompt incorporating all user selections, level calibration, and source-style instructions.

3

AI Generation

Claude (claude-sonnet) processes the prompt and returns a structured JSON array of questions, each with type, marks, answer, and optional examiner tip.

4

Rendering

The interface parses the JSON and renders questions with toggleable answers, type badges, and mark allocations.

5

Export

User downloads a formatted PDF with question paper on page 1 and mark scheme on page 2, including metadata header and total marks.

Technology Stack

Layer	Technology
Frontend UI	React (JSX) with Tailwind-style CSS variables
AI Engine	Anthropic Claude API — claude-sonnet-4
PDF Generation	Client-side HTML→PDF (browser print) or ReportLab (server-side)
Knowledge Style	Prompt engineering referencing PMT, SME, Corbettmaths patterns
Export	Blob download for HTML/PDF; future: Google Drive, Classroom API

Phase 1 — Core (Now)

- Topic + subject + board input • 5 difficulty levels (Foundation → University) • 7 question types with mix control • 1–30 question slider
- 8 knowledge base toggles • Instant mark scheme generation • PDF export (question paper + mark scheme) • Show / hide answers per question

Phase 2 — Engagement (Next 3 months)

- Timed mock mode with countdown timer • Score yourself — tick correct / incorrect + auto score • Question bank — save and reload sets
- Mixed topic mode — combine multiple topics • Examiner tip toggle per question • Word (.docx) export alongside PDF

Phase 3 — Intelligence (3–6 months)

- Syllabus mapping to exam-board specification points • Adaptive difficulty within a session • Performance tracker across sessions
- Spaced repetition scheduler • AI-powered wrong-answer explanations • Formula sheet auto-attachment for sciences

Phase 4 — Platform (6–12 months)

- Teacher dashboard — create and share sets to classes • Google Classroom / Microsoft Teams integration • Student accounts with progress history
- Public question bank (community-shared sets) • Mobile app (iOS / Android) • School / institution licensing

The AI Question Generator solves a real, daily pain point for thousands of students and teachers. By combining intelligent AI generation with trusted knowledge-base styles, flexible customisation, and instant professional PDF export, it collapses hours of exam prep into seconds.

Who it's for	GCSE and A-Level students, sixth-form and secondary school teachers, private tutors, revision clubs.
Core value	Save 30–60 minutes per revision session. Get exam-quality questions in the right style, at the right level, instantly.
Unique edge	Multi-source knowledge bases, real exam-board alignment, and mark scheme generation in one place — no competitor does all three.
Growth path	Starts as a personal tool, scales to a teacher platform and eventually a school-licensed SaaS product.

Built with Claude AI · Designed for UK exam preparation · Version 1.0